# ■ Security Analysis Report

Generated: 2026-02-13 14:28:36

## Summary

| Metric | Value |
|---|---|
| Total Vulnerabilities | 16 |
| Security Score | 23/100 |
| Critical | 3 |
| High | 7 |
| Medium | 6 |
| Low | 0 |

## Vulnerabilities

### *[1] Taint Analysis: SQL Injection*

| | |
|---|---|
| **Severity** | Critical |
| **Category** | Data Flow Vulnerability |
| **File** | test_samples\vulnerable.php |
| **Line** | 7 |
| **Rule ID** | TAINT-SQL-INJECTION |

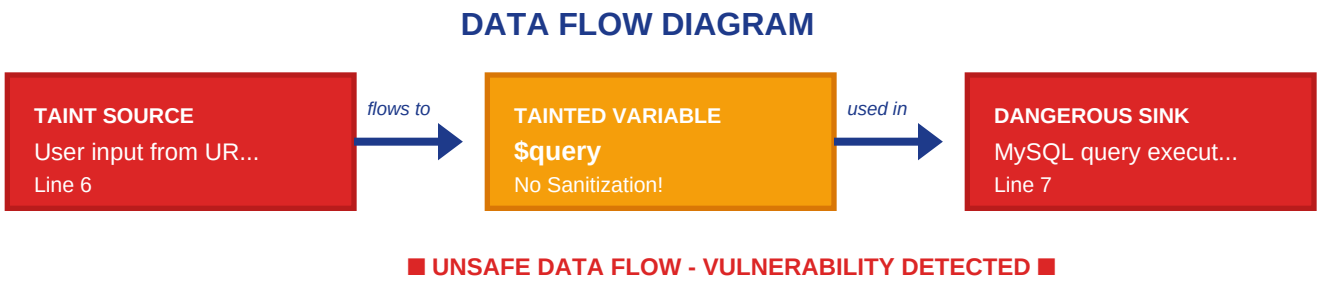
**Description:** Tainted data from User input from URL parameters (line 6) flows through variable 'query' to MySQL query execution

**Code:**

```
$query = "SELECT * FROM users WHERE id = " . $userId; $result = mysqli_query($conn, $query); // A03: XSS
```

**Remediation:** Sanitize data from 'User input from URL parameters' before using in 'MySQL query execution'. Data flow: Source: User input from URL parameters -> Propagated to query

**Data Flow Diagram:**




## *[2] Taint Analysis: Command Injection*

| | |
|---|---|
| **Severity** | Critical |
| **Category** | Data Flow Vulnerability |
| **File** | test_samples\vulnerable.php |
| **Line** | 23 |
| **Rule ID** | TAINT-COMMAND-INJECTION |

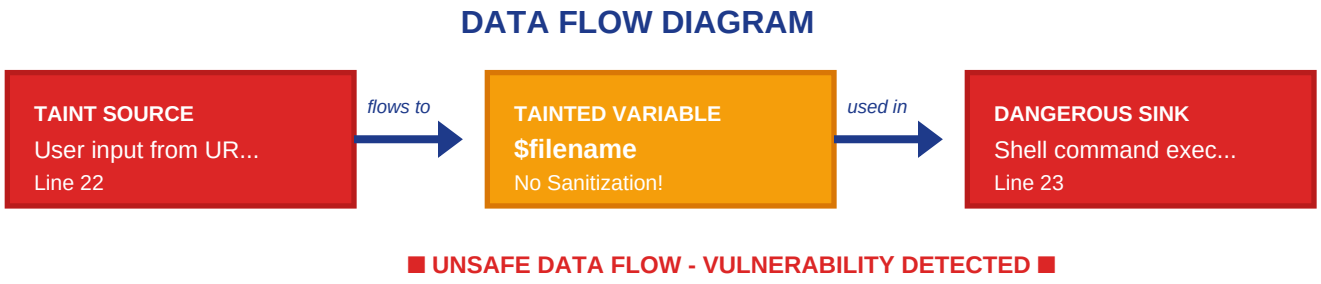
**Description:** Tainted data from User input from URL parameters (line 22) flows through variable 'filename' to Shell command execution

**Code:**

```
$filename = $_GET['file']; exec("cat " . $filename); // A10: SSRF
```

**Remediation:** Sanitize data from 'User input from URL parameters' before using in 'Shell command execution'. Data flow: Source: User input from URL parameters

**Data Flow Diagram:**



### *[3] Taint Analysis: Path Traversal*

| | |
|---|---|
| **Severity** | High |
| **Category** | Data Flow Vulnerability |
| **File** | test_samples\vulnerable.php |
| **Line** | 27 |
| **Rule ID** | TAINT-PATH-TRAVERSAL |

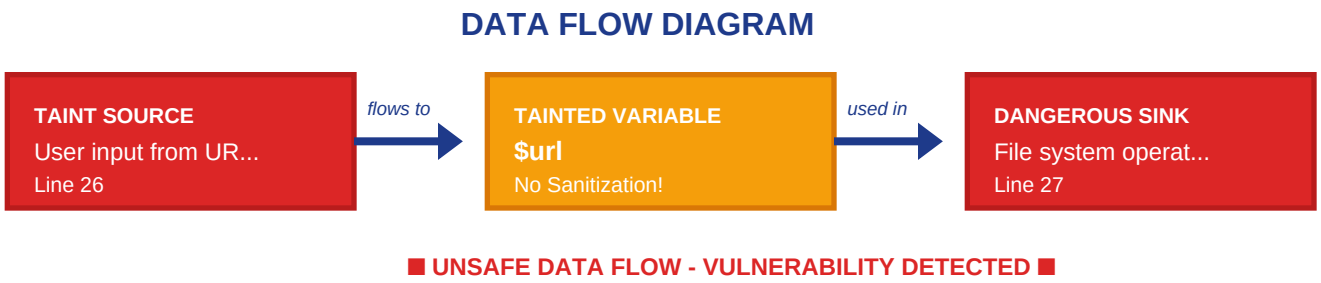
**Description:** Tainted data from User input from URL parameters (line 26) flows through variable 'url' to File system operation

**Code:**

```
$url = $_GET['url']; $content = file_get_contents($url); // A07: Weak Session
Management
```

**Remediation:** Sanitize data from 'User input from URL parameters' before using in 'File system operation'. Data flow: Source: User input from URL parameters

**Data Flow Diagram:**




### *[4] Taint Analysis: Server-Side Request Forgery*

| | |
|---|---|
| **Severity** | High |
| **Category** | Data Flow Vulnerability |
| **File** | test_samples\vulnerable.php |
| **Line** | 27 |
| **Rule ID** | TAINT-SERVER-SIDE-REQUEST-FORGERY |

**Description:** Tainted data from User input from URL parameters (line 26) flows through variable 'url' to Remote file access

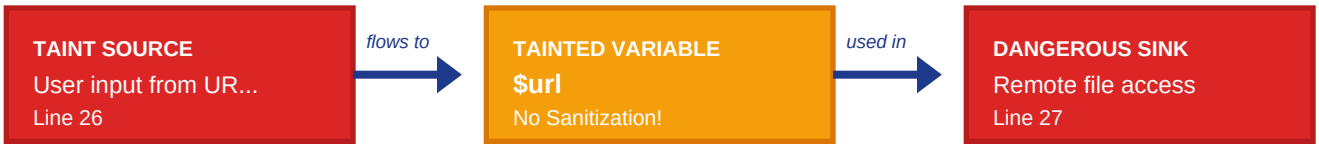
**Code:**

```
$url = $_GET['url']; $content = file_get_contents($url); // A07: Weak Session
Management
```

**Remediation:** Sanitize data from 'User input from URL parameters' before using in 'Remote file access'. Data flow: Source: User input from URL parameters

**Data Flow Diagram:**

## DATA FLOW DIAGRAM




**■ UNSAFE DATA FLOW - VULNERABILITY DETECTED ■**

### *[5] Taint Analysis: Path Traversal*

| | |
|---|---|
| **Severity** | High |
| **Category** | Data Flow Vulnerability |
| **File** | test_samples\vulnerable.php |
| **Line** | 27 |
| **Rule ID** | TAINT-PATH-TRAVERSAL |

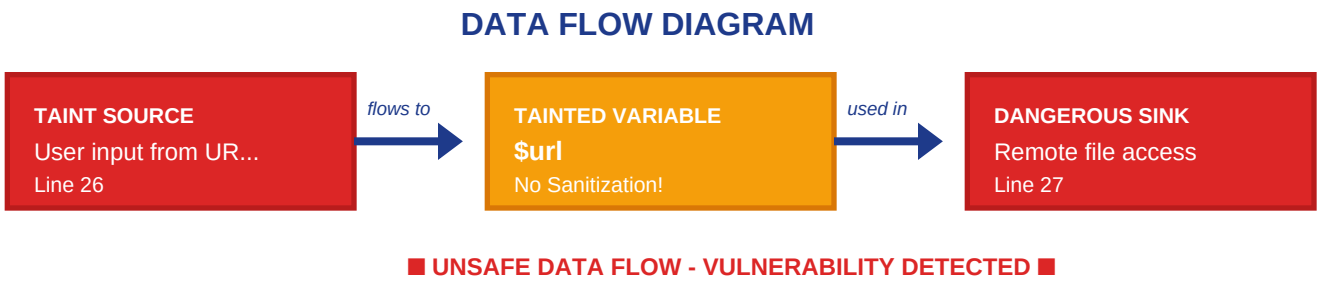
**Description:** Tainted data from User input from URL parameters (line 26) flows through variable 'url' to Remote file access

**Code:**

```
$url = $_GET['url']; $content = file_get_contents($url); // A07: Weak Session
Management
```

**Remediation:** Sanitize data from 'User input from URL parameters' before using in 'Remote file access'. Data flow: Source: User input from URL parameters

**Data Flow Diagram:**




### *[6] Missing Input Validation*

| | |
|---|---|
| **Severity** | Medium |
| **Category** | Best Practice |
| **File** | test_samples\vulnerable.php |
| **Line** | 5 |
| **Rule ID** | BP-PHP-001 |

**Description:** Direct use of superglobals without validation

**Code:**

```
// A03: SQL Injection $userId = $_GET['id']; $query = "SELECT * FROM users WHERE id = "
. $userId; $result = mysqli_query($conn, $query);
```

**Remediation:** Validate and sanitize all input using filter_input() or filter_var()

### *[7] Missing Input Validation*

| | |
|---|---|
| **Severity** | Medium |
| **Category** | Best Practice |
| **File** | test_samples\vulnerable.php |
| **Line** | 10 |
| **Rule ID** | BP-PHP-001 |

**Description:** Direct use of superglobals without validation

**Code:**

```
// A03: XSS echo $_GET['name']; print $_POST['comment'];
```

**Remediation:** Validate and sanitize all input using filter_input() or filter_var()

### *[8] Missing Input Validation*

| | |
|---|---|
| **Severity** | Medium |
| **Category** | Best Practice |
| **File** | test_samples\vulnerable.php |
| **Line** | 11 |
| **Rule ID** | BP-PHP-001 |

**Description:** Direct use of superglobals without validation

**Code:**

```
echo $_GET['name']; print $_POST['comment']; // A02: Weak Password Hashing
```

**Remediation:** Validate and sanitize all input using filter_input() or filter_var()

### *[9] Missing Input Validation*

| | |
|---|---|
| **Severity** | Medium |
| **Category** | Best Practice |
| **File** | test_samples\vulnerable.php |
| **Line** | 14 |
| **Rule ID** | BP-PHP-001 |

**Description:** Direct use of superglobals without validation

**Code:**

```
// A02: Weak Password Hashing $password = $_POST['password']; $hash = md5($password);
```

**Remediation:** Validate and sanitize all input using filter_input() or filter_var()

### *[10] Weak Password Hashing*

| | |
|---|---|
| **Severity** | Critical |
| **Category** | A02 |
| **File** | test_samples\vulnerable.php |
| **Line** | 15 |
| **Rule ID** | A02-PHP-001 |

**Description:** Use of weak cryptographic hash functions

**Code:**

```
$password = $_POST['password']; $hash = md5($password); // A05: Hardcoded Secrets
```

**Remediation:** Use password_hash() with PASSWORD_BCRYPT or PASSWORD_ARGON2ID

### *[11] Hardcoded Secrets*

| | |
|---|---|
| **Severity** | High |
| **Category** | A05 |
| **File** | test_samples\vulnerable.php |
| **Line** | 18 |
| **Rule ID** | A05-PHP-001 |

**Description:** Hardcoded credentials or secrets in source code

**Code:**

```
// A05: Hardcoded Secrets $password = "admin123"; $api_key =
"sk_live_1234567890abcdef";
```

**Remediation:** Use environment variables or secure configuration files outside web root

### *[12] Hardcoded Secrets*

| | |
|---|---|
| **Severity** | High |
| **Category** | A05 |
| **File** | test_samples\vulnerable.php |
| **Line** | 19 |
| **Rule ID** | A05-PHP-001 |

**Description:** Hardcoded credentials or secrets in source code

**Code:**

```
$password = "admin123"; $api_key = "sk_live_1234567890abcdef"; // A03: Command
Injection
```

**Remediation:** Use environment variables or secure configuration files outside web root

### *[13] Missing Input Validation*

| | |
|---|---|
| **Severity** | Medium |
| **Category** | Best Practice |
| **File** | test_samples\vulnerable.php |
| **Line** | 22 |
| **Rule ID** | BP-PHP-001 |

**Description:** Direct use of superglobals without validation

**Code:**

```
// A03: Command Injection $filename = $_GET['file']; exec("cat " . $filename);
```

**Remediation:** Validate and sanitize all input using filter_input() or filter_var()

### *[14] Missing Input Validation*

| | |
|---|---|
| **Severity** | Medium |
| **Category** | Best Practice |
| **File** | test_samples\vulnerable.php |
| **Line** | 26 |
| **Rule ID** | BP-PHP-001 |

**Description:** Direct use of superglobals without validation

**Code:**

```
// A10: SSRF $url = $_GET['url']; $content = file_get_contents($url);
```

**Remediation:** Validate and sanitize all input using filter_input() or filter_var()

### *[15] Server-Side Request Forgery*

| | |
|---|---|
| **Severity** | High |
| **Category** | A10 |
| **File** | test_samples\vulnerable.php |
| **Line** | 27 |
| **Rule ID** | A10-PHP-001 |

**Description:** File/URL access with user-controlled input, vulnerable to SSRF

**Code:**

```
$url = $_GET['url']; $content = file_get_contents($url); // A07: Weak Session
Management
```

**Remediation:** Validate URLs, whitelist allowed domains, block internal network access

### *[16] Weak Session Management*

| | |
|---|---|
| **Severity** | High |
| **Category** | A07 |
| **File** | test_samples\vulnerable.php |
| **Line** | 30 |
| **Rule ID** | A07-PHP-001 |

**Description:** Missing session security configuration

**Code:**

```
// A07: Weak Session Management session_start(); // Missing session security
configuration
```

**Remediation:** Configure secure session settings: httpOnly, secure flag, SameSite attribute, regenerate session ID on login